

Deeksha Ramakrishna

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PROFESSIONAL SUMMARY

AI & Cloud Engineer with 4+ years of experience designing, building, and deploying scalable AI and machine learning solutions to drive business impact. Strong expertise in the full AI lifecycle, traditional ML, and LLM-powered systems, including RAG pipelines and Agentic AI workflows. Proven ability to translate complex business challenges into performant, reliable technical solutions while optimizing systems for latency, cost, scalability, and reliability.

TECHNICAL SKILLS

Languages & Databases: Python, SQL, Bash, MongoDB, MySQL, MariaDB, PostgreSQL

AI/ML: Supervised & Unsupervised Learning, Reinforcement Learning, Data Engineering, Deep Learning, Predictive Modelling, Fine-Tuning, Agentic AI, RAG, LLMs, NLP, Generative AI, MLOps, Prompt Engineering, Document Intelligence(OCR)

Frameworks & Tools: TensorFlow, Keras, PyTorch, Scikit-Learn, LangChain, LangGraph, HuggingFace, Pinecone, MLFlow, KubeFlow, SageMaker, Vertex AI, Spark, Kafka, Airflow, NLTK, JAX, XGBoost, OpenAI, Claude, Pandas, Matplotlib, Numpy, REST APIs, FastAPI, Flask

Cloud, DevOps & SRE: AWS, GCP, OpenStack, Kubernetes, Docker, Terraform, CloudFormation, Ansible, Helm, Argo CD, Linux Administration and Troubleshooting, Prometheus, Grafana, CI/CD (GitHub Actions, gitlab, Jenkins), GitOps

PROFESSIONAL EXPERIENCE

AI and Cloud Infrastructure Engineer, Elef Capital (Algorithmic Trading Startup), Atlanta, GA Jan 2026 - Present

- Owned the end-to-end design and deployment of an **ML-driven strategy and sentiment analysis pipeline** to AWS, operating with high autonomy in a fast-paced startup environment.
- Supported deployment workflows by containerizing ML workloads with Docker and orchestrating via Kubernetes to **zero-downtime execution and high availability**.
- Designed reliable feature and inference pipelines, maintaining robust **CI/CD pipelines via GitHub Actions** to enable scalable MLOps practices.
- Defined and maintained **SLIs/SLOs and error budgets for critical services**, enhancing system observability through Prometheus and Grafana to proactively detect issues.

Machine Learning Engineer | Teaching Assistant (TA), UMD, College Park, MD Feb 2024 - Dec 2025

- Developed **RAG pipelines and Agentic AI workflows** to build an automated research proposal review system utilizing Apache Kafka and Airflow, translating complex federal compliance challenges into an AI solution that **reduced compliance check times by 90%**.
- Built and deployed a production-grade **NLP translation pipeline leveraging Apache Spark**, achieving a **BLEU score of 44** to enhance cross-lingual research.
- As a TA**, I directed technical training for 60+ engineers on **Docker and Kubernetes orchestration** enforcing best practices in version control and containerized applications.

Site Reliability Engineer, SysEleven GmbH, Berlin, Germany Jul 2021 - Dec 2023

- Managed cloud resources and **maintained hybrid multi-cloud infrastructure** (OpenStack, AWS) using Infrastructure as Code tools including Terraform and Ansible and also configured and managed Linux-based KVM/QEMU hypervisors.
- Pioneered the company's first ML initiative** by designing and deploying an end-to-end automated anomaly detection system for platform logs, **achieving 92% precision** leveraging ETL pipeline, model training and evaluation.
- Provisioned and maintained bare-metal servers** and OpenStack configurations; performed core infrastructure operations including system patching, OS upgrades, automated backups, and capacity planning.
- Expanded provider-level observability via Prometheus queries and Grafana dashboards/alerts; **tracked SLIs/SLOs, reducing MTTR by 25%**.
- Troubleshooting network issues such as DNS failures, TCP/IP packet analysis, and routing configurations using standard Linux diagnostic tools.
- Participated in on-call rotations and blameless post-mortems; wrote **Python/Bash automation** for operational tooling and maintained **platform-wide CI/CD pipelines** using gitlab to reduce toil.

EDUCATION

Master of Engineering in Cloud Engineering - University of Maryland, College Park, MD, USA

Master of Science in Data & Knowledge Engineering (Applied ML) - Otto Von Guericke University, Germany

Bachelor of Engineering in Computer Science and Engineering - Dr. Ambedkar Institute of Technology, India